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Longitudinal Growth of Mandated Hearing Aid Benefits in the US

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IMPORTANCE Inconsistent hearing aid coverage across US health plans poses substantial barriers to essential care. Prior research has examined private insurance and Medicaid mandates separately, leaving the overall reach of coverage across all major US programs unclear.

OBJECTIVE To identify mandated hearing aid coverage under private insurance, Medicaid, and federal direct-service programs and to compare coverage for US youth, adults, and older adults by coverage plan and economic indicators.

DESIGN, SETTING, AND PARTICIPANTS This cross-sectional study used population-level policy surveillance data with nationally representative insurance data from the American Community Survey (ACS) from 2008 to 2023. ACS respondents were categorized by age (younger than 18 years, 18 to 64 years, and 65 years or older), insurance type (state-level private insurance, Medicaid, and federal direct-service insurance providers), and eligibility for hearing aid coverage. Data were analyzed from February 2024 to August 2026.

EXPOSURES Mandated health coverage for hearing aids through private insurers, Medicaid, US Department of Veterans Affairs, TRICARE, and Indian Health Service.

MAIN OUTCOMES AND MEASURES Annual estimates of the proportion of US individuals with mandated coverage for hearing aids across insurance types, classified into coverage groups. ACS measures included are age, annual household income, and home ownership status.

RESULTS Among 50 203 993 ACS respondents from 2008 to 2023, mandated hearing aid coverage increased from 34.1% (95% CI, 34.0-34.2) to 52.5% (95% CI, 52.4-52.6) among youth younger than 18 years. Among adults aged 18 to 64 years, coverage rose from 10.2% (95% CI, 10.1-10.2) to 19.2% (95% CI, 19.2-19.3), with Medicaid the largest contributor. Mandated coverage for adults 65 years and older declined from 18.5% (95% CI, 18.4-18.6) to 16.9% (95% CI, 16.8-17.0), with Medicaid the primary source. Mandated hearing aid coverage grew 1.03 percentage points (95% CI, 1.02-1.04; $P < .001$) faster per year among youth younger than 18 years than among adults 65 years and older, largely reflecting consistent Medicaid coverage through the long-standing Early and Periodic Screening, Diagnostic, and Treatment (EPSDT) benefit and the adoption of private insurance pediatric mandates. In 2023, renters and individuals in households below the federal poverty level threshold were more likely to hold mandated coverage than homeowners and those at or above the threshold in every age group.

CONCLUSIONS AND RELEVANCE In this cross-sectional study, despite gradual improvements, substantial gaps remained in mandated hearing aid coverage, particularly for older adults and adults with limited economic resources. Policy reforms, including expanded Medicaid coverage and mandates requiring hearing aid benefits in private insurance and Medicare, could improve access and health outcomes.

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Hearing loss (HL) interventions enhance communication, social engagement, and quality of life, yet hearing aids are not widely used.[1-6] HL has consequences across the life course: for children and young adults, unaided HL affects language development, educational attainment, and employment opportunities,[7,8] while in older adults, it is associated with increased risk for falls, social isolation, and dementia.[9-11] In the US, hearing aid affordability is a leading barrier to care.[12,13] Mean out-of-pocket costs for prescription hearing aids range from $2000 to $7000 per set,[14] and coverage often excludes the cost of evidence-based clinical services required for optimal use. Hearing aid coverage, defined here as insurance-provided payments for costs of device provision and associated fitting and maintenance services provided by an audiologist or hearing instrument specialist, is also inconsistent across private and public health insurance programs.[1,3,5,6,12] Legislative and policy efforts have sought to reduce affordability barriers by requiring hearing aid benefits in some insurance and service delivery programs. However, these efforts have developed unevenly across coverage sources, making it difficult to assess the overall reach of hearing aid coverage over time.[15]

Insurance coverage is one access pathway to financial support for hearing aids and related care. Hearing aid coverage for adults is distributed nonuniformly across private insurance, public programs, and federal direct-service insurance providers, like the US Department of Veterans Affairs (VA), TRICARE, and the Indian Health Service (IHS). Each operates under different legal authorities, with different eligible populations and mandating mechanisms (**Table 1**; eAppendix 2 in Supplement 1). The result is a fragmented system shaped by the policies of individual insurance and service delivery programs.

Laws (eg, statutes and regulations) sometimes require health plans to include a prescription hearing aid benefit, which this study refers to as *mandates*. Existing research has focused primarily on state-level private insurance mandates or Medicaid policies,[1-6] leaving other major coverage sources underexamined. As a result, policymakers and clinicians cannot currently identify which populations remain most exposed to hearing aid affordability barriers, because, to our knowledge, no analysis has spanned private insurance, Medicaid, and federal direct-service programs together. This consideration is important because hearing care needs differ by age, while the ability to absorb out-of-pocket hearing aid costs may vary substantially by household income. How has population exposure to mandated hearing aid coverage changed across major US insurance and direct-service programs, and how does this vary by age and household income? We address these gaps by conducting an analysis of mandated prescription hearing aid coverage (including related services) across major US insurance and direct-service programs from 2008 to 2023. Building on previously published state-level datasets of private insurance mandates and Medicaid policies, this study adds original policy data on federal direct-service programs and integrates these into a longitudinal cross-program analysis estimating the share of US individuals with mandated hearing aid coverage overall, by coverage source, and across age and economic indicator groups.

Key Points

Question How has mandated hearing aid coverage in the US changed from 2008 to 2023 by age and economic indicators?

Findings In this cross-sectional study including 50 203 993 American Community Survey respondents, mandated coverage rose among youth younger than 18 years (34.1% to 52.5%) and among working-age adults aged 18 to 64 years (10.2% to 19.2%) but declined modestly among adults 65 years and older (18.5% to 16.9%). Most adults in households below the federal poverty level threshold lacked guaranteed coverage.

Meaning Mandated hearing aid coverage grew substantially for youth and working-age adults but not older adults, and most adults living in households with low incomes did not have guaranteed benefits.

Methods

This cross-sectional study combined legal epidemiology methods with data from the American Community Survey (ACS) to conduct a trend analysis of mandated hearing aid coverage policies across state-level private insurance, Medicaid, and federal direct-service insurance providers (Table 1) from 2008 to 2023. Details on the legal data sources, coding procedures, and ACS variables are provided below, and a comprehensive description of hearing aid policy data collection, coding decisions, population weighting, and sensitivity analyses is provided in eAppendix 2 in Supplement 1. The University of South Florida Institutional Review Board deemed this work exempt from human subjects oversight. Briefly, we classified mandated hearing aid coverage using 2 main sources: (1) original policy data on federal direct-service programs and (2) previously published state-level datasets of private insurance mandates and Medicaid hearing aid policies.[2,4] We then linked coded policy indicators to insurance coverage and demographic data to estimate the annual US population exposure to mandated hearing aid coverage across the 15-year study period. Race and ethnicity data are self-reported in the ACS; we categorized race data as American Indian or Alaska Native, Asian, Black, White, and other race (including not elsewhere classified and multiple races). Ethnicity choices in the survey were Hispanic or non-Hispanic.

We focus on mandated coverage rather than observed benefit uptake because mandates can be measured consistently across heterogeneous systems. Realized access to hearing aid benefits, in contrast, reflects policy requirements, plan design and selection, and benefit awareness, which are not consistently measured across insurance types. For example, hearing aid benefits in Medicare Advantage (MA) and private plans may vary substantially in generosity and are not directly observable in national survey data. A mandate-based framework therefore provides a standardized, policy-relevant measure of whether individuals are exposed to hearing aid coverage requirements, while recognizing that beneficiaries—particularly older adults, among whom MA enrollment has grown substantially—may also access hearing aids through nonmandated benefits.

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Table 1. Sources of Insurance for Hearing Health Coverage

Program	Coverage source	Mandate determination authority	Eligible population	Historic mandated hearing coverage	Mandated hearing coverage scope
Insurance plans					
Group private, fully insured	Employer-sponsored health insurance purchased from private insurers	Insurers (within state and federal regulation); employers select plan options	Employees and dependents	Varies by jurisdiction and time period	Services vary; includes hearing aids and hearing evaluations
Nongroup private, self-insured	Individually purchased private insurance; employer self-funded plan	Insurers; federal government (ACA); state government; employer if ERISA or self-funded	Individuals and families; employees of large firms	Varies by jurisdiction and time period	Services vary; includes hearing aids and hearing evaluations
Medicaid (youth <18 y)	Joint federal-state program, publicly financed	Federal government (EPSDT); states administer within federal rules	Eligible youth from birth to age 20 y	Coverage since 1989	Comprehensive pediatric hearing services required, including hearing aids when medically necessary
Medicaid (adults aged 18-64 y)	Joint federal-state program, publicly financed	State government and program determinations; some federal government baseline requirements	Low-income; categorical eligibility	Varies by jurisdiction and time period	Services vary by state; may include hearing evaluations, hearing aids, and related services
Medicare Part A/B (traditional)	Federal program, publicly financed	Federal government (CMS)	Adults ≥65 y and eligible individuals with disabilities	Statutory exclusion for hearing aids in place since 1965	Diagnostic hearing examinations
Medicare Part C (Medicare Advantage)	Privately administered plans funded through federal Medicare payments	Private insurers (within CMS regulations)	Adults ≥65 y who are eligible or individuals with disabilities who elect Medicare Advantage	Variable since 2000s; hearing aid benefits optional and plan specific	Services vary by plan; may include hearing examinations, hearing aids, and related services as supplemental benefits
Federal direct-service providers					
VA	Federal program	Federal government (US Department of Veterans Affairs)	Veterans	Since the 1950s	Covers comprehensive hearing services including hearing aids when indicated
IHS[a]	Federal program	Federal government (US Indian Health Service)	Eligible American Indian and Alaska Native persons	Varies by jurisdiction and time period	Varies by service unit; may include hearing evaluations and hearing aids
TRICARE[b]	Federal program	Federal government (US Department of Defense)	Active-duty military, retirees, and dependents	Since 2001	Covers diagnostic hearing services; hearing aids covered for active duty and certain dependents with specified medical need

Abbreviations: ACA, Patient Protection and Affordable Care Act; CMS, US Centers for Medicare & Medicaid Services; EPSDT, Early and Periodic Screening, Diagnostic, and Treatment; ERISA, Employee Retirement Income Security Act of 1974; IHS, Indian Health Service; VA, US Department of Veterans Affairs.

[a] IHS operates in conjunction with tribal hearing programs.

[b] TRICARE is administered through private contractors with access to direct federal care.

Data on private and Medicaid hearing aid coverage—including effective dates of laws and policies and eligibility criteria—were extracted from datasets developed by Arnold et al.[2,4] Because these prior datasets did not include military or veteran coverage, we conducted original policy surveillance for VA and TRICARE using legal epidemiology methods (**Table 2**).[16-34] We also reviewed federal policy governing the IHS. Although hearing services may be provided through the IHS, coverage is not established by federal mandate; it depends on available appropriations, facility capacity, and independent tribal programming.[35-41] Accordingly, IHS was classified as a nonmandate program in this analysis.

Policy data were merged with ACS data from 2008 (the first year for which insurance type became available) to 2023 (the most recent year available), extracted from the Integrated Public Use Microdata Series USA (IPUMS USA).[42] The ACS, conducted

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Table 2. Federal Direct-Service Provider Hearing Coverage Policies for Military Personnel and Dependents

Policy feature	VA	TRICARE
Eligibility	• Veterans with service-connected hearing disabilities (conductive, mixed, or sensory)[16-20] • Veterans with non–service-related hearing conditions if eligibility criteria are met[16-19] • Veterans with profound deafness or those requiring rehabilitation devices[20-22]	• Active-duty service members[19-21] • Active-duty service members' dependents with statutorily defined profound hearing loss: ≥40 dB HL in 1 or both ears at designated frequencies; ≥26 dB HL in ≥3 frequencies; speech recognition score <94%[23,24]
Covered hearing services	• Hearing aids and replacements if damaged or destroyed[20,24,25] • Fitting and training for prosthetic appliances, including institutional training[20,24-26] • Devices to support rehabilitation (eg, telecaptioning devices)[26,27]	• Hearing aids and related audiology services [23,24,28,29]
Costs	• No out-of-pocket costs for hearing aids, accessories, aural rehabilitation, tinnitus management devices, or related services[a] • No spending caps; includes premium technology levels from all major manufacturers	• Minimal or no out-of-pocket costs for covered services
Alignment with audiological best practices	• Policies highly aligned with current evidence-based clinical audiology standards; limitations mainly related to eligibility	• Benefits designed to ensure access to high-quality audiology consistent with clinical best practices
Historical coverage	• Coverage documented since the 1950s (rooted in post–World War II and post–Vietnam War needs, when audiology and hearing aid provision originated in the VA)[30-32]	• Benefits implemented since at least December 2001[28,29]

Abbreviations: HL, hearing loss; VA, US Department of Veterans Affairs.

[a] VA priority groups (1-8) rank veterans by service-connected disability rating, income, and special circumstances, with lower numbers indicating higher priority and greater benefits. Although VA provides hearing aids, repairs, and batteries at no charge to eligible veterans, associated audiology visits may carry copays for veterans in priority groups 7 and 8, whose hearing loss is not connected to service, with amounts varying by income level.

annually by the US Census Bureau, is a nationally representative, repeated cross-sectional survey providing individual-level data on age, health insurance type, state of residence, and survey year. Respondents may report multiple insurance types, including employer-provided or union-provided private insurance, directly purchased private insurance, TRICARE or other military insurance, Medicaid, Medicare, VA health care, IHS, or no coverage.[42]

Respondents were classified as having hearing aid coverage if they met age eligibility requirements in states with applicable Medicaid or private insurance mandates or if they reported VA or TRICARE coverage (among working-age adults aged 18 to 64 years). Hearing aid benefits were not considered mandated for Medicare, IHS, or self-insured private plans due to variability or absence of coverage requirements. For Medicare, this reflects the statutory exclusion of hearing aids and the optional, plan-specific nature of benefits under MA.

The ACS does not distinguish self-insured from fully insured private plans, and self-insured plans are not subject to state-level mandates under the Employee Retirement Income Security Act of 1974.[43] Thus, we followed prior methods[4] and applied state-by-year estimates from the Medical Expenditure Panel Survey[44] to estimate the probability that privately insured respondents were in fully insured (covered by a state-level hearing aid mandate) or self-insured (not covered by a state-level hearing aid mandate) plans. Beneficiaries eligible for both Medicare and Medicaid (dual eligible) were identified using IPUMS USA assignment rules, with Medicaid mandates applied when respondents met state age criteria.

When respondents reported multiple insurance types, we assigned them to mutually exclusive groups according to the most generous hearing aid benefits available, on the rationale that individuals would use coverage where mandates applied (eAppendix 1 in Supplement 1). The priority ordering differs by age group and is detailed in eFigure in Supplement 1. This prioritization reflects prior findings that private insurance mandates cover a wider age range and specify more generous benefits compared with Medicaid.[2,4,45] For states with midyear policy changes, affected respondents contributed to the categories relevant to them for the fraction of the year in effect. For population estimates (eTable 1 in Supplement 1), respondents with probabilistic exposure contributed fractionally to coverage categories; for analyses using a single binary coverage status, each such respondent was instead assigned to 1 category by a single random draw against that probability (eAppendix 1 in Supplement 1).

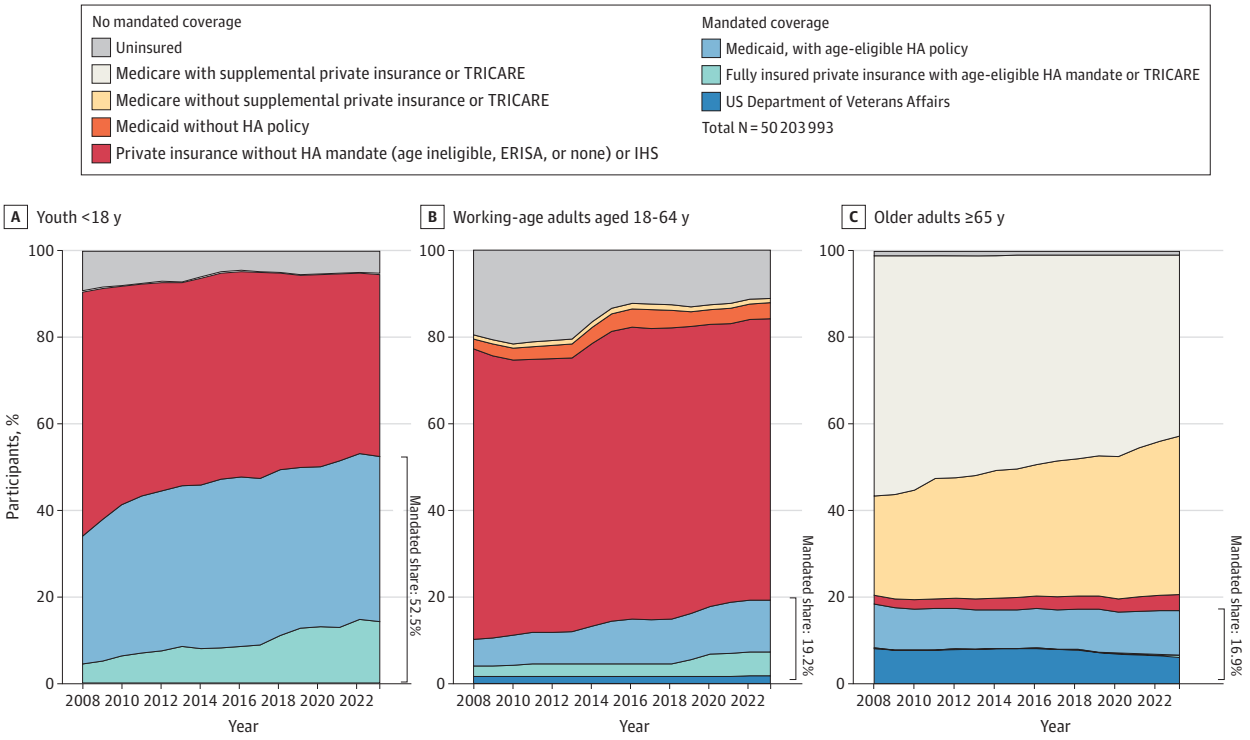
Following coverage group assignment, ACS person-level survey weights were applied to derive estimates of the population share belonging to each coverage group for each year. From this, we summed groups with mandated coverage to identify the national share at any point with mandated coverage. We explored this separately for 3 age groups: youth (younger than 18 years), working-age adults (aged 18 to 64 years), and older adults (65 years and older). Annual changes in mandated hearing aid coverage were estimated with an individual-level linear probability model on pooled ACS data (2008 to 2023), regressing an indicator for mandated coverage on a linear year term, controlling for state fixed effects and individual-level controls of race and ethnicity, employment status, and educational attainment. Age group × year interactions estimated youth and working-age group differences in annual percentage point growth relative to the older adult group. Estimates were weighted using ACS survey weights, with design-based standard errors accounting for the ACS sampling design (Taylor-series linearization). We also described economic indicators (poverty status based on the federal poverty level [FPL] threshold and housing tenure, defined as owner occupied vs renter occupied) by mandated hearing aid coverage status within each age group for 2023.

Analyses were conducted using Stata version 18.0 (StataCorp). *t* Tests were used to generate 95% CIs for coverage rates and regression estimates, and χ^2 tests were used to test dif-

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Figure. Health Insurance Coverage Rates by Age and Mandated Hearing Aid Coverage



Health insurance coverage rates by hearing aid coverage status across all 50 states and Washington, DC, from 2008 through 2023, stratified by age group: youth younger than 18 years, working-age adults aged 18 to 64 years, and older adults 65 years and older. Data sources include the Integrated Public Use Microdata Series American Community Survey (2008-2023); our database of state-level Medicaid hearing aid policies; our database of state-level private insurance hearing aid mandates[4]; and the Medical Expenditure Panel Survey state-year estimates for percentage of private-sector enrollees enrolled in self-insured plans at establishments that offer health insurance from 2008 to 2023. ERISA indicates Employee Retirement Income Security Act of 1974; HA, hearing aid; IHS, Indian Health Service.

ferences in means. All *P* values were 2-tailed, and significance was set at $P < .05$. Data were analyzed from February 2024 to August 2026.

Results

This study included 50 203 993 ACS respondents from 2008 to 2023. The **Figure**, A, describes trends in insurance type and mandated hearing aid coverage status for youth younger than 18 years from 2008 to 2023; insurance types with mandated coverage are grouped together and those without mandated coverage are grouped to allow for aggregation. Medicaid, including Early and Periodic Screening, Diagnostic, and Treatment (EPSDT) coverage, was the largest contributor, increasing from 29.6% (95% CI, 29.5-29.7) to 38.1% (95% CI, 38.0-38.3). Overall mandated coverage for this age group rose from 34.1% (95% CI, 34.0-34.2) to 52.5% (95% CI, 52.4-52.6). The Figure, B, shows the same trends for working-age adults aged 18 to 64 years. The overall percentage with mandated coverage in this age group was substantially lower than for youth, increasing from 10.2% (95% CI, 10.1-10.2) to 19.2% (95% CI, 19.2-19.3). Medicaid again was the largest source of mandated hearing aid coverage, followed by private insurance and VA. Most youth and working-age adults without mandated coverage had private insurance. Finally, the Figure, C, displays trends for older adults 65 years and older, who expectedly reported predominantly Medicare insurance and experienced minor declines in mandated hearing aid coverage rates over time (18.5% [95% CI, 18.4-18.6] to 16.9% [95% CI, 16.8-17.0]). Medicaid was the largest contributor of mandated coverage, accounting for 10.4% (95% CI, 10.3-10.4) of adults 65 years and older in 2023, followed by VA and private insurance.

Table 3 presents regression estimates of annual coverage growth by age group. Mandated coverage grew approximately 0.66 percentage points (95% CI, 0.66-0.67; $P < .001$) faster per year for working-age adults aged 18 to 64 years than older adults 65 years and older and 1.03 percentage points (95% CI, 1.02-1.04; $P < .001$) faster per year for youth younger than 18 years than older adults . Coverage also grew 0.36 percentage points (95% CI, 0.35-0.37; $P < .001$) faster per year for youth than for working-age adults.

Table 4 presents annual household income (stratified by FPL threshold) and home ownership status (stratified by housing unit tenure) of ACS respondents by mandated hearing aid coverage status in 2023. Within every age group, individuals in households below the FPL threshold were more likely to have mandated coverage than those at or above the threshold, consistent with Medicaid's central role as a mandate pathway. The

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Table 3. Regression Estimates of Mandated Hearing Aid Coverage by Age Group, 2008-2023[a]

Variable	β (95% CI)	SE	*P* value
Year since 2008	0.001 (−0.005 to 0.007)	0.003	.80
Age group			
Youth (<18 y)	9.793 (9.652 to 9.934)	0.072	<.001
Working-age adult (18-64 y)	−1.883 (−1.953 to −1.813)	0.036	<.001
Older adult (≥65 y)	1 [Reference]	NA	NA
Age group × year			
Youth × year	1.030 (1.020 to 1.040)	0.005	<.001
Working-age adult × year	0.662 (0.655 to 0.670)	0.004	<.001
Older adult × year	1 [Reference]	NA	NA
Total ACS respondents, No.	50 203 993	NA	NA

Abbreviations: ACS, American Community Survey; NA, not applicable.

[a] Linear probability model on individual-level ACS data pooled across 2008 to 2023; the outcome is an indicator for mandated hearing aid coverage scaled to percentage points. Year enters linearly with 2008 coded as 0; the reference age group is adults 65 years and older, so age group × year coefficients give each group's difference in annual growth relative to adults 65 years and older. Model includes individual-level controls of state, employment (employed, unemployed, not in labor force, or missing), education (less than high school, high school, some college, bachelor or more, or missing), recoded self-reported Hispanic indicator (Hispanic or non-Hispanic), and recoded self-reported race (American Indian or Alaska Native, Asian, Black, White, or other race [including not elsewhere classified and multiple races]). Survey robust standard errors are used. Estimates are weighted using ACS-designated survey weights. Sources included the Integrated Public Use Microdata Series USA American Community Survey (2008-2023), our databases of Medicaid and private insurance hearing aid policies, and the Medical Expenditure Panel Survey state-year estimates of self-insured enrollment (2008-2023).

same pattern was evident for renters relative to homeowners in all age groups (Table 4). Nevertheless, most adults reporting annual household incomes below the FPL threshold lacked mandated hearing aid coverage in 2023, including 63.5% (95% CI, 63.3-63.7) of working-age adults aged 18 to 64 years and 71.3% (95% CI, 70.9-71.6) of older adults 65 years and older in households below the FPL threshold.

Discussion

This study provides a longitudinal cross-program assessment of mandated hearing aid coverage in the US from 2008 to 2023. The strengths of this study include its longitudinal scope and breadth of coverage; by integrating original policy data on federal direct-service programs with previously published Medicaid and private insurance policy databases and linking these to 15 years of nationally representative ACS data, it provides a population-weighted, cross-program analysis of mandated hearing aid coverage not previously available in the literature. The inclusion of federal direct-service insurance providers alongside state-level Medicaid and private insurance mandates addresses an important gap in prior work, which has examined programs largely in isolation. To our knowledge, no independent published study has estimated population-level exposure to mandated hearing aid coverage across these programs nationally, precluding direct comparison of coverage estimates with an external national benchmark.

We found that exposure to mandated coverage increased over time, but growth differed substantially by coverage source and age group. Importantly, this analysis provides evidence of substantial growth in mandated coverage among youth and working-age adults. This growth is consistent with legislative and policy efforts meaningfully increasing guaranteed hearing aid coverage for many, suggesting that coverage requirements are associated with expanded population-level access to financial support for hearing devices and related services.

At the same time, coverage expansion was less evident among older adults, for whom hearing care needs and affordability barriers may be especially pronounced. eTable 3 in Supplement 1 appears to attribute a decline to a lower share of the veteran population that would benefit from the VA's generous hearing aid coverage. The Figure, C, shows a decline in Medicare plus supplemental (private or TRICARE) coverage and an increase in Medicare without supplemental coverage, suggesting that fewer older adults had an additional coverage pathway captured in our mandate framework. This pattern should be interpreted in the context of rapid MA growth and the increasing availability of supplemental hearing benefits, as 94% of MA plans now report offering hearing benefits.[46] However, these benefits are plan specific rather than guaranteed by law and vary substantially in dollar limits, network access, device types, and cost sharing.[46-48] Thus, our estimates may understate older adults' exposure to some forms of hearing aid coverage. The observed decline among older adults should be interpreted as a reflection of the mandate framework's structural limitations given the concurrent growth of nonmandated, optional MA benefits. It is important to note that, despite the availability of optional hearing aid coverage under MA, only 8% of MA enrollees reported using hearing services annually.[49] Even with this caveat, our mandate-based findings identify persistent gaps in guaranteed coverage.

Mandated hearing aid coverage was lowest among older adults. In the absence of guaranteed coverage, people with HL remain responsible for substantial out-of-pocket costs or forgoing care.[50] Uninsured individuals face the greatest exposure, as they remain outside both mandated and plan-based hearing aid coverage pathways. The program-specific findings help explain these age-related patterns. Throughout the study period, private insurance plans without hearing aid man-

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Table 4. Socioeconomic Characteristics of American Community Survey (ACS) Respondents by Mandated Hearing Aid Coverage Status, 2023[a]

	ACS respondents, % (95% CI)					
	Youth <18 y (n = 635 622)		Working-age adults aged 18-64 y (n = 1 847 807)		Older adults ≥65 y (n = 746 415)	
Variable	No mandate	With mandate	No mandate	With mandate	No mandate	With mandate
Below the official federal poverty level threshold	16.6 (16.3-16.8)	83.4 (83.2-83.7)	63.5 (63.3-63.7)	36.5 (36.3-36.7)	71.3 (70.9-71.6)	28.7 (28.4-29.1)
At or above the official federal poverty level threshold	55.7 (55.5-55.8)	44.3 (44.2-44.5)	83.9 (83.9-84.0)	16.1 (16.0-16.2)	86.2 (86.1-86.2)	13.8 (13.8-13.9)
Renter	31.0 (30.8-31.2)	69.0 (68.8-69.2)	75.5 (75.4-75.6)	24.5 (24.4-24.6)	72.9 (72.6-73.1)	27.1 (26.9-27.4)
Homeowner	57.4 (57.2-57.5)	42.6 (42.5-42.8)	84.3 (84.2-84.4)	15.7 (15.6-15.8)	86.7 (86.6-86.8)	13.3 (13.2-13.4)

[a] Percentages are weighted using ACS survey weights; numbers are unweighted respondent counts. Given the very large samples, conventional hypothesis tests are not informative here; we therefore emphasize the magnitude of differences (row percentages) rather than test statistics. Federal poverty level threshold defined by the US Census Bureau for 2023; poverty status is based on household income and the poverty threshold for each respondent's household size. Home ownership tenure was defined at the level of the housing unit and represents residence in an owner-occupied vs renter-occupied housing unit. There were 175 965 missing responses for both poverty and homeownership status in the 2023 survey. Sources include the Integrated Public Use Microdata Series USA American Community Survey (2023) and our databases of Medicaid and private insurance hearing aid policies.

dates accounted for a large share of both youth and working-age adults without guaranteed coverage. For Medicaid-insured youth, the longstanding federal EPSDT requirement[51] provides a consistent coverage pathway for medically necessary services, including hearing aids.

In contrast, adult Medicaid hearing aid coverage varies widely by state (eTable 2 in Supplement 1),[2,45] contributing to inconsistent access among working-age and older adults. Growth in mandated coverage among adults is consistent with contributions from the adoption of Medicaid hearing aid policies, increases in Medicaid enrollment (including through Medicaid expansion under the Patient Protection and Affordable Care Act), and the growing number of state private insurance hearing aid mandates, although this analysis cannot isolate the independent contribution of each factor. These changes represent progress toward broader access, but coverage remains incomplete because adult benefits continue to hinge on state of residence and insurance type. The larger annual growth in mandated coverage among youth relative to working-age adults also raises concerns about continuity of coverage as individuals age out of pediatric eligibility pathways. Youth may have access to hearing aid coverage through EPSDT or pediatric private insurance mandates, but those protections often do not continue into adulthood, creating the potential for abrupt loss of coverage even when hearing care needs persist.

Beyond age-related differences in coverage pathways, our findings clarify how mandated coverage is distributed by household income and homeownership. Because Medicaid is the dominant mandate pathway, more individuals reporting annual household income below the FPL threshold held mandated coverage than individuals with incomes at or above the FPL threshold in every age group, and the same was true of renters relative to homeowners. Prior research documents a disproportionate out-of-pocket cost burden for hearing aid purchases among individuals reporting low income.[13,52,53] The absence of guaranteed coverage carries the greatest financial consequence for those least able to absorb it: nearly two-thirds of working-age adults and more than 70% of older adults below the FPL threshold held no mandated hearing aid benefit in 2023. Moreover, because a large proportion of mandated coverage operates largely through Medicaid, access for eligible adults still depends on insurance provider participation, benefit awareness, and state-specific benefit scope. Future analyses should examine the state-specific effects of Medicaid expansion status and the coverage gap on hearing aid access, particularly for adults with limited economic resources in states that have not expanded Medicaid eligibility.

Our study highlights several policy pathways for expanding mandated hearing health coverage. A federal mandate requiring all insurance plans to cover hearing aids would be the broadest lever, improving access for youth and working-age adults currently enrolled in private plans without mandates. Expanding Medicaid coverage federally by requiring hearing aid benefits for all adult beneficiaries, not only those dually eligible for Medicare and Medicaid coverage, would benefit adults with low incomes across age groups, including working-age adults in states that have not yet adopted adult hearing aid coverage and older adults dually enrolled in Medicare and Medicaid. Reducing uninsurance would also produce meaningful coverage gains for both youth and working-age adults. The stability of VA and TRICARE coverage also illustrates the value of a durable, federally supported pathway and provides a useful policy model for reducing variability seen across state-dependent and plan-dependent coverage systems. Finally, policy reforms targeting Medicare specifically—including requiring Medicare Part B to cover hearing aids and strengthening benefit floors for hearing services under MA—would address the persistent coverage gap among older adults, for whom no other mandate pathway currently provides reliable access.

Limitations

This study has limitations. This study focuses on mandated hearing aid benefits because they represent the policy levers that define population-level access to coverage. Accordingly, our estimates are not intended to measure the share of individuals with any hearing aid coverage. Coverage may exist outside of mandates through voluntary private insurance benefits, optional MA

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offerings, or noninsurance programs, such as Georgia's Hearing Aid Distribution Program[54] or the US Department of Defense Retiree-At-Cost Hearing Aid Program,[55] which are not captured in this analysis. This study focuses on coverage mandates for prescription hearing aids and did not separately identify or code provisions specific to over-the-counter hearing aids, which became available in 2022.[15] Future research should examine whether and how over-the-counter hearing aids are addressed within existing coverage policies. Our study has other limitations related to measurement and scope. Individuals reporting IHS coverage were classified as not having hearing aid coverage in the absence of a secondary plan, which may underestimate access among tribal populations, as some tribes independently provide hearing services.[35-38] Additionally, ACS does not distinguish TRICARE beneficiaries by status (eg, active duty, retiree, or dependent), despite differences in eligibility for hearing aid benefits, which may underestimate coverage among eligible subgroups.[56] Accordingly, estimates involving IHS and TRICARE should be interpreted with caution. Finally, we documented large gains in mandated benefits among working-age adults aged 18 to 64 years that were mainly attributable to growth in Medicaid coverage; however, it was not possible to disentangle the impacts of specific policy or population factors, such as state hearing aid coverage policies, Medicaid eligibility expansions, and changes in population demographic characteristics.

Conclusions

HL is a highly prevalent and disruptive chronic condition affecting individuals in the US, yet coverage for hearing aids, a primary effective intervention, remains among the least guaranteed. The gradual progress observed in this study from 2008 to 2023 demonstrates that policy change can expand access to hearing aids. Closing the remaining gaps, particularly for older adults and those in states without private insurance mandates, will require deliberate policy action to ensure that access to hearing care is determined by clinical need, not geography or insurance type. Aligning hearing aid coverage policy with clinical need through consistent mandates and expanded benefits can improve health outcomes and quality of life.[57-61]

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